# SHEET # 4

4.1 Define the following items: Time invariant discrete – time system, Linear discrete – time system, and causal signal.

4.2 Find and plot the z-transform of the following discrete-time sequences:

(a) $x(n)=\{0,1,3,5,3,1,0\}$

(b) $x(-3)=0$, $x(-2)=1$, $x(-1)=3$, $x(0)=5$, $x(1)=3$, $x(2)=1$ & $x(3)=0$.

4.3 Find and plot the z-transform of each of the following discrete-time sequences:

(a) $x(n) = \sin(nT)$, $n=0,1,\ldots$

(b) $x(n) = a^n$, $n > 0$

$= 0$ $n < 0$

4.4 An exponential sequence is defined as

$$x(n) = e^{-kn}, \quad n \geq 0$$

Find and plot its z-transform, including the constraint on the z-transform to converge, for: (a) k is real. (b) k is complex.

4.5 Find, using the power series method the first five values of causal discrete-time sequence corresponding to each of the following z-transform:

(a) $X(z) = (z - 1)/(z - 0.7071)^2$

(b) $X(z) = 1/(z - 0.5)(z + 0.9)^3$

4.6 Repeat problem (5) using the partial fraction expansion method.

4.7 Repeat problem (5) using the residue method.

4.8 Given the difference equation

$$y(n) + B_1 y(n-1) + B_2 y(n-2) = A$$

where A, $B_1$ and $B_2$ are arbitrary constants, find an expression for z-transform, Y(z). Use an appropriate inverse z-transform technique to obtain a closed form for y (n).

4.9 For the discrete-time system shown in Fig.4.13, obtain the difference equation, and then derive the system transfer function H (z).

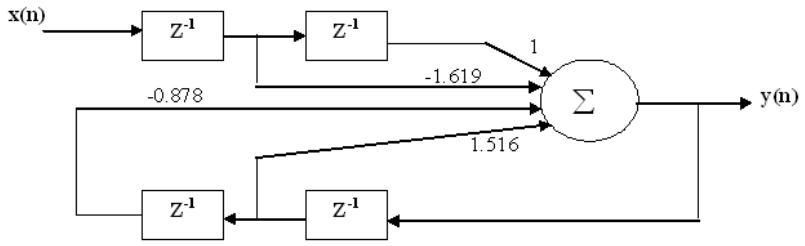



Fig.4.13

4.10 The transfer function of a discrete-time system has poles at z=0.5, z=0.1±j0.02 and a third-order zero at z=-1 and z=1.

(a) Sketch the pole-zero diagram for the system.

(b) Derive the system transfer function, H (z), from the pole-zero diagram.

(c) Develop the difference equation.

(d) Draw the realization diagram in signal flowgraph form.

4.11 The signal flow graph of a discrete-time system is shown in Fig. 4.14 Find the two-step difference equation relating the output, y (n), and input, x (n). Derive the transfer function, H (z), from the difference equation.

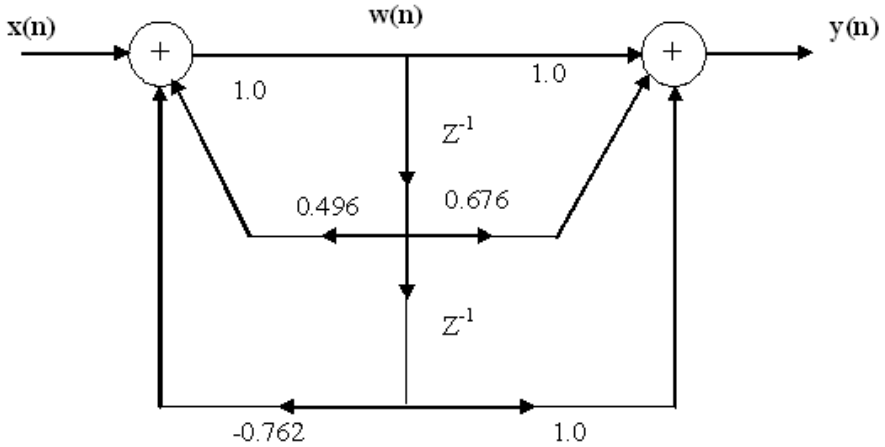



Fig.4.14

# Assignment # 4

A4.1 Find and plot the z-transform of the following discrete-time sequences:

(i) $x(n) = \{0,1,3, \pm 5,-3,-1,0\}$

(ii) $x(n) = 1, \quad 0 \leq n \leq N-1$ (use N=10)

$= 0, \quad$ elsewhere

A4.2 Find, using the power series method the first five values of causal discrete-time sequence corresponding to the following z-transform:

$$X(z) = (z^3 - z^2 + z - 1)/(z + 0.9)^3$$

A4.3 Repeat problem (A4.2) using the partial fraction expansion method.

A4.4 Repeat problem (A4.2) using the residue method.